**Step 2:** The product of common factors is 2 x 2 and also the product of the factors that are not common is 3 x 13 x 31.

**Step 3:** Now, LCM = product of common factors x product of factors that are not common

= (2 x 2) x (3 x 13 x 31) = 4 x 1209 = 4836

Thus, LCM of 156 and 124 is 4836.

**(or)**

156 = 2 x 78 = 2 x 2 x 39 = 2 x 2 x 3 x 13;

124 = 2 x 62 = 2 x 2 x 31

The prime factor 2 appears a maximum of 2 times in the prime factorization of 156 and 124, the prime factor 3 appears only 1 time in the prime factorization of 156, the prime factor 13 appears only 1 time in the prime factorization of 156 and the prime factor 31 appears only 1 time in the prime factorization of 124.

Hence, the required LCM = (2 x 2) x 3 x 13 x 31 = 4836.

## 1.7 Application Problems on HCF and LCM

Let us see the word problems that involve the HCF and the LCM concepts in daily life situations.

**Example 6:** What is the greatest number that will divide 62, 78 and 109 leaving remainders 2, 3 and 4 respectively?

**Solution:** Get all the common factors of 62 – 2, 78 – 3 and 109 – 4, i.e., 60, 75 and 105 and see that the common factors will divide them all. The greatest number is the H.C.F of 60, 75 and 105.

60=2 x 2 x 3 x 5  75=3 x 5 x 5  105=3 x 5 x 7

Hence, the HCF is 3 x 5=15, which is the greatest number that will divide 62, 78, 109 leaving remainders 2, 3 and 4 respectively.

**Example 7:** A book seller has 175 English books, 245 Science books and 385 Mathematics books. He wants to sell the books in a box, subject-wise in equal numbers. What will be the greatest number of the boxes required? Also find the number of books for each subject in a box.

**Solution:** This is a HCF related problem. So, we need to find the HCF of 175, 245 and 385.

| | |
|---|---|
| 5 | 175 |
| 5 | 35 |
| 7 | 7 |
| | 1 |

| | |
|---|---|
| 5 | 245 |
| 7 | 49 |
| 7 | 7 |
| | 1 |

| | |
|---|---|
| 5 | 385 |
| 7 | 77 |
| 11 | 11 |
| | 1 |

175 = 5 x 5 x 7;  245 = 5 x 7 x 7;  385 = 5 x 7 x 11



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